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MILITARY SPECIFICATION

REELS, FUELING HOSE, MANUALLY OPERATED

1. SCOPE

1.1 Scope.- This specification covers manually operated fueling hose reels.

1.2 Classification.- Hose reels shall be of either of the following types, as specified (see 6.1):

Type I - For 150 feet of collapsible hose of 2-1/2 inch size.
Type II - For 150 feet of noncollapsible hose of 1-1/2 inch size.
Type III - For 150 feet of noncollapsible hose of 1-1/2 inch size or for 150 feet of collapsible hose of 2-1/2 inch size.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-620 - Aluminum Bronze, Plate, Bar, Sheet and Strip.
QQ-B-679 - Bronze, Aluminum; Rod, Flat Products with Finished Edges (Flat Wire, Strip, and Bar), Shapes and Forgings.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-963 - Drawings; Electrical Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals).
MIL-P-15929 - Primer Coating, Shipboard, Vinyl-Red Lead (Formula No. 119 - for Hot Spray).
MIL-P-15930 - Primer Coating, Shipboard, Vinyl-Zinc Chromate (Formula No. 120 - for Hot Spray).
MIL-B-16033 - Bronze, Aluminum; Castings.
MIL-E-16366 - Electrical Clamps, Lug Terminals and Conductor Splices — Pressure Grip.
MIL-H-17902 - Hose Assemblies, Rubber, Aircraft Fueling, Collapsible Type.
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-P-20693 - Plastic Molding Material, Polyamide (Nylon), Rigid.
MIL-H-21291 - Hose Assemblies, Synthetic Rubber, Aviation Fuel, Suction and Discharge, Noncollapsible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4930

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Qualification.- The hose reels furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Materials.- Except as otherwise specified herein, the hose reels shall be constructed of aluminum, aluminum bronze, or other suitable nonsparking materials, all as approved by the bureau or agency concerned. Unless otherwise specified in the contract or order, all fittings conveying gasoline through the hose reel shall be made of aluminum bronze in accordance with class 2 of MIL-B-16033 or QQ-A-620 and QQ-B-679. In order to prevent deterioration in sea atmosphere, aluminum parts should not contact brass or bronze; however, where such aluminum parts come in contact with brass or bronze, they shall be protected by a permanent coating in accordance with MIL-P-15328 and MIL-P-15929. Materials shall be suitable for the operating range of temperature specified in 3.5. All steel parts shall be cadmium plated in accordance with QQ-P-416, after which 2 coats of zinc chromate primer shall be applied. All aluminum parts shall be given a coat of zinc chromate primer. Zinc chromate primer shall be in accordance with either TT-P-645 or MIL-P-15930. The contact spider shown (see figures 2 and 3) shall be molded nylon type I of MIL-P-20693 and the assembly shall be in accordance with the requirements of MIL-H-17902 or MIL-H-21291, as applicable.

3.3 Construction.- The hose reel shall be of a durable, rigid construction, as light in weight and compact as possible. The drum assembly (drum, reel sides and its reinforcements) shall withstand the hose pull without distortion or collapse and shall be capable of withstanding some sideways hose pull without undue distortions. The construction of the completed drum assembly shall be such that disassembly can be readily accomplished with standard tools. When bolted down to a true flat surface the shaft and rotating seal center lines shall be in true alignment and the drum assembly shall turn without binding under a rim pull not to exceed 7 pounds with the reel unpressurized. Both ends of the drum shaft shall be supported on ball or roller bearings which shall have provision for easy lubrication. Unless otherwise specified, these reels shall be capable of containing three 50 foot lengths of hose, as required, for types I, II, or III, with consideration given to normal reeling practices. The hose used will be either collapsible (see MIL-H-17902) or noncollapsible (see MIL-H-21291), as applicable. The drum diameter shall not be less than the minimum bending diameter of the hose. The general outline and base mounting dimensions and weight shall be in accordance with figure 1.

3.4 Reels shall be capable of passing 200 gallons per minute (g.p.m.) of aviation fuel at a pressure of 150 pounds per square inch (p.s.i.) and shall be capable of withstanding a static fuel pressure of 225 p.s.i. without leakage or other signs of weakness. Restrictions and turbulence through the reel shall be kept to a minimum. For this purpose, the fluid path shall be of the 3 inch size (including rotating swivel joint) and shall utilize long sweep elbows. The complete fluid handling section of the reel (that is, from inlet flange to flanged adapter) shall be capable of being removed as a unit without disassembly of the drum or reel sides.

3.5 Reels shall be of a weathertight design with particular care taken that all bearings or other corrodible parts are adequately protected from the weather. Provision shall be made for lubrication of working parts such as bearings or other mating surfaces and all such parts shall be lubricated prior to delivery with a lubricant soluble in gasoline. The reels and the lubricant used shall be capable of satisfactory operation within the range of temperatures from minus 20° to plus 120°F. Bearing seals shall be such that the pressure due to forced lubrication will not bind the shaft.

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3.6 Support.- The reel drum assembly shall be supported at two points, one on each side of the drum assembly. A single cantilever support is not acceptable.

3.7 Reel drum.- The reel drum may have either smoothly formed flat sides or suitable spoke ribs, so formed as not to damage the hose. The rims of the drum shall provide a good grip for hand rewind.

3.8 Holding brake.- Reels shall be provided with a holding brake to lock the reel in any position. In addition, the reel shall be fitted with a suitable device for clamping the hose nozzle securely to the reel to prevent unwinding of the hose and damage to the nozzle.

3.9 Reel hub discharge.- The reel hub discharge shall be so arranged as to provide a smooth second layer of hose essentially as shown on figures 1 and 2. The hub discharge fitting and adapter shall be in accordance with figures 2 and 3 for types I and II, respectively. Reels shall be right or left hand operation and top or bottom wind, as specified (see 6.1). A right hand, top wind reel is defined as one which, when looking into the flanged adapter on the discharge fitting when it is at the top of the drum, has its inlet connection on the right hand side. (Note that a top wind, right hand reel and a bottom wind, left hand reel are identical.) It shall be possible by turning the discharge fitting around or by other simple means, to enable the hose to unreel from the bottom of the reel drum to either side of the reel (that is, to make the reel either right hand or left hand) without rotating the reel drum.

3.10 Discharge fitting.- Unless otherwise specified herein, the discharge fitting shall have an electrical contact at the center to bear against a similar contact button in the hose. The electrical contact spider shall be in accordance with MIL-H-17902 or MIL-H-21291, as applicable (see 3.2). From this contact on the discharge fitting, there shall be a continuous electrical path through the center of the hose reel to a terminal on the stationary part of the reel at the inlet connection (see figures 1 and 2). The electrical circuit through the reel shall be electrically insulated from the reel. The stationary terminal shall be in accordance with MIL-E-16366. Electrical contact button on the discharge fitting shall be as shown on figures 2 and 3.

3.11 Gasoline tight rotating joint.- The reel shall have a gasoline tight rotating joint between reel inlet connection and the rotating hub on the reel drum, the design shall be satisfactory to the bureau or agency concerned. The rotating joint shall not bind under loading (or shall be so supported to prevent a bending load being applied), shall have provision for lubrication of moving parts, and shall be capable of easy disassembly for maintenance purposes. The rotating joint shall be tight under a vacuum of 25 inches Hg. as well as the pressure specified in 3.4.

3.12 Inlet connection.- The reel inlet connection shall be flanged, the flange being part of the rotating swivel joint. Unless otherwise specified in the contract or order, the flange thickness and bolt hole drilling shall be in accordance with class 250 of MIL-F-20042. The bolt holes shall straddle the vertical centerline of the flange.

3.13 Drawings.-

3.13.1 Preliminary drawings.- Within 30 days after award of contract, the contractor shall submit to the bureau or agency concerned preliminary drawings in accordance with 3.13.2, that are sufficient to permit evaluation of the design and approval of materials (see 3.2). These drawings shall be detail assembly drawings and shall show the following:

(a) Overall views, sectional views and dimensions to delineate the hose reel and the materials of the parts sufficient to provide Naval activities with the necessary information to install, repair and maintain the reel.
(b) Space required for removal of parts with the reel installed.
(c) The operating conditions and capacity.
(d) Location of all connections and mounting holes, electrical, etc.
(e) Test approval data.
(f) Weight.
(g) Any special installation requirements.
(h) Special tools required for servicing.

3.13.1.1 If a contractor proposes to furnish equipment which previously has been supplied and for which drawings have been previously approved, resubmission of drawings for approval will not be required. Reference to the approved drawing, or the furnishing of a copy of the approved drawing (where required by the bureau or agency concerned not having the drawing on record) will be acceptable.

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3.13.2 Production drawings.- When specified (see 6.1), detail assembly drawings shall be submitted to the bureau or agency concerned by the contractor for completion of the contract or order. These drawings shall be in accordance with MIL-D-963, except for the extent of detail. Only the information required on the preliminary drawing need be furnished (see 3.13.1). The cognizant inspector shall validate production drawings after receipt of approved preliminary drawing.

3.14 Screw threads.- All screw threads shall conform to Handbook H28.

3.15 Identification of product.- Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130. The size of the hose to be used shall be stamped on each reel.

3.16 Workmanship.- Workmanship shall be first class in every respect and parts shall be free of all imperfections which may impair serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2. Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.2.1, 4.4, and 4.5. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

4.2.1 Shock tests.- The class HI shock tests shall be grade A, class I, type A tests in accordance with MIL-S-901. The reel shall be completely assembled (including hose), and shall be mounted on the shock machine in a manner simulating a typical shipboard installation. The number of blows shall be as specified in MIL-S-901.

4.2.1.1 Examination after shock tests.- Upon completion of the shock test the equipment shall be carefully examined to determine the extent of any damage to the mechanical components. Examination of the reels shall determine the following:

(a) The hose shall be unreeled and then replaced on the reel several times. Any undue noise or difficulty in operation shall be investigated and the cause determined.
(b) Any unbalance shall be measured and shall not exceed twice the amplitude obtained prior to shock.
(c) A conductivity check shall be made to insure that the electrical circuit through the reel is intact.
(d) The fluid handling section of the reel (between inlet flange and discharge connection) shall be given a hydrostatic test to 200 p.s.i. for strength and porosity and a kerosene or test fluid (solvent) test to 50 p.s.i. and 120 p.s.i. for tightness without showing any signs of leakage or other weakness.
(e) The reels shall be disassembled following the examination after shock tests specified and examined thoroughly for damage. The effects of the shock tests and subsequent check tests on the structure shall be carefully observed and recorded.

4.2.1.2 Failure to pass shock tests.- The hose reel shall be considered to have failed to pass the shock tests in the event of the following:

(a) Breakage of any parts including mounting bolts.
(b) Appreciable distortion or dislocation of any parts, including mounting feet, bearings.
(c) Excessive unbalance determined as specified in 4.2.1.1.
(d) Failure to pass the hydrostatic or electrical conductivity tests.

4.3 Sampling.-

4.3.1 Lot.- All reels of the same type presented at one time shall be considered a lot for quality conformance inspection. For lot acceptance tests and visual examination of any components, parts, or subassemblies, lots shall be formed and the number of items in any lot shall be not less than one-fourth of the contract quantity.

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4.3.2 Sampling for examination and tests.- Sample reels shall be selected in accordance with Inspection Level II of MIL-STD-105 and the acceptable quality level equal 1.5 percent defective for the examination and tests specified in 4.4 and 4.5.

4.4 Examination.- Sample reels selected in accordance with 4.3.2 shall be examined to determine conformance with all the requirements of this specification, not involving tests. Examination shall be conducted as specified in table I.

Table I - Classification of defects in accordance with MIL-STD-105.

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type, not as specified.
102	Hose reel incomplete; parts missing or improperly assembled.
103	Materials defective or not as specified; evidence of cracks, deep pits, blowholes, shrinkage defects or foreign inclusions.
104	Bearings defective or not protected from the weather.
105	Reel not double supported.
106	Reel drum sides not smoothly finished or drum rims do not provide a comfortable hand grip.
107	Brake for locking drum in any position missing or defective.
108	Not provided with a means of clamping the hose nozzle securely to the reel.
109	Discharge fitting cannot be turned around or not provided with a simple means of reversing the fitting to make the reel either right or left hand operation.
110	Electrical contact buttons missing, defective, not securely mounted and insulated, or contacts not as specified.
111	Reel inlet connection not the required size or flange not smoothly finished.
112	Hub discharge adapter missing, not securely mounted, or not as specified; threads stripped, torn, crossed shrunken (not concentric) not perfect full threads, or thread type and size not as specified.
113	Bolt hole diameter or spacing or inlet flange not as specified, or thickness of flange less than required.
114	Width of drum not within the specified tolerance.
115	Steel parts not cadmium plated, dissimilar metals that come in contact with each other not coated as required, or aluminum parts not coated as required.
116	Reel wobbles, sticks, binds, or evidences excessive looseness, amount of rim pull required to rotate reel exceeds the specified maximum.
Minor:	
201	Mounting dimensions (bolt hole diameter or spacing) not as specified.
202	Diameter of reel exceeds the maximum permissible.
203	Overall height ("A" dimension) exceeds the maximum permissible.
204	Overall width ("B" dimension) exceeds the maximum permissible.
205	Horizontal or vertical location inlet flange ("C" and "F" dimensions) not within the specified tolerance.
206	Stationary electric terminal defective or not as specified.
207	Weight of reel exceeds the maximum allowable.
208	Marking, manufacturer's name or trademark missing, illegible, incorrect or not as specified.

4.4.1 Defects.- Any defective reels in the sample shall not be offered for delivery. If the number of defective reels in any sample exceeds the acceptance number, this shall be cause for rejection of the entire lot.

4.5 Tests.-

4.5.1 Operating test.- Assembled reels selected in accordance with 4.3.2 shall be loaded with hose and operated by hand to test all functions, fits, clearances and adjustments. Hose for the test will be furnished by the Government on a loan basis.

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4.5.2 Hydrostatic test.- In addition to the test specified in 4.5.1, the fluid-carrying portion of the hose shall be hydrostatically tested to 200 p.s.i. for strength and porosity.

4.5.2.1 Strength and porosity.-

4.5.2.2 Tightness.- The reel shall also be hydrostatically tested by kerosene or synthetic test fluid (solvent) to 50 p.s.i. and 120 p.s.i. for tightness.

4.5.3 Leakage.- The reel shall show no leakage or any other signs of weakness after the performance of the tests specified in 4.5.1 and 4.5.2.

4.5.4 Chemical analysis.- At the discretion of the inspector the chemical analysis shall be performed by the manufacturer to assure the use of materials conforming to the manufacturer's approved drawings.

4.5.5 Possible test failures.- Possible test failures of the reels shall be determined as follows:

(a) Operating test:
  (1) Failure of reel to contain the specified length of hose, evidence of bending, wobbiing or binding when loaded reel is rotated.
(b) Hydrostatic test:
  (1) Evidence of leakage or binding of the seals when the indicated hydrostatic pressure is applied to the fluid-carrying portion of the reel.
(c) Kerosene or synthetic test fluid (solvent):
  (1) Evidence of leakage or binding of the seals when the specified fluid pressure is applied to the fluid-carrying portion of the reel.
(d) Chemical analysis (if required):
  (1) Chemical composition of component parts materials not as specified.

4.6 Inspection of preservation, packaging, and packing.- Sample items and packages shall be selected and inspected in accordance with MIL-P-116 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Reels.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be as specified (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

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5.2.1.1 Preservation and packaging.-

5.2.1.1.1 Level A.- The reels shall be packaged in accordance with method III of MIL-P-116. All openings shall be sealed with blank flanges, threaded plugs or tape conforming to type I or II, class 1 of PPP-T-60, as applicable.

5.2.1.1.2 Level C.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2.1.2 Packing.-

5.2.1.2.1 Level A.- The reels shall be individually packed in overseas type containers conforming to PPP-B-621, PPP-B-601 or PPP-B-585. The boxes shall be modified by the addition of a nominal 2 by 4 inch skid placed flat. Requirements for closure and strapping shall be in accordance with the appendix to the applicable specification. The reels shall be shrouded within the containers with barrier material conforming to UU-P-271.

5.2.1.2.2 Level B.- The reels shall be individually packed in domestic type containers conforming to PPP-B-621, PPP-B-601, PPP-B-585 or PPP-B-591. The boxes shall be modified by the addition of nominal 2 by 4 inch skids placed flat. Requirements for closure and strapping shall be in accordance with the appendix to the applicable specification.

5.2.1.2.3 Level C.- The reels shall be packed to insure carrier acceptance and safe delivery to destination. Containers shall comply with the Uniform Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2.1.3 Marking.- In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with MIL-STD-129.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

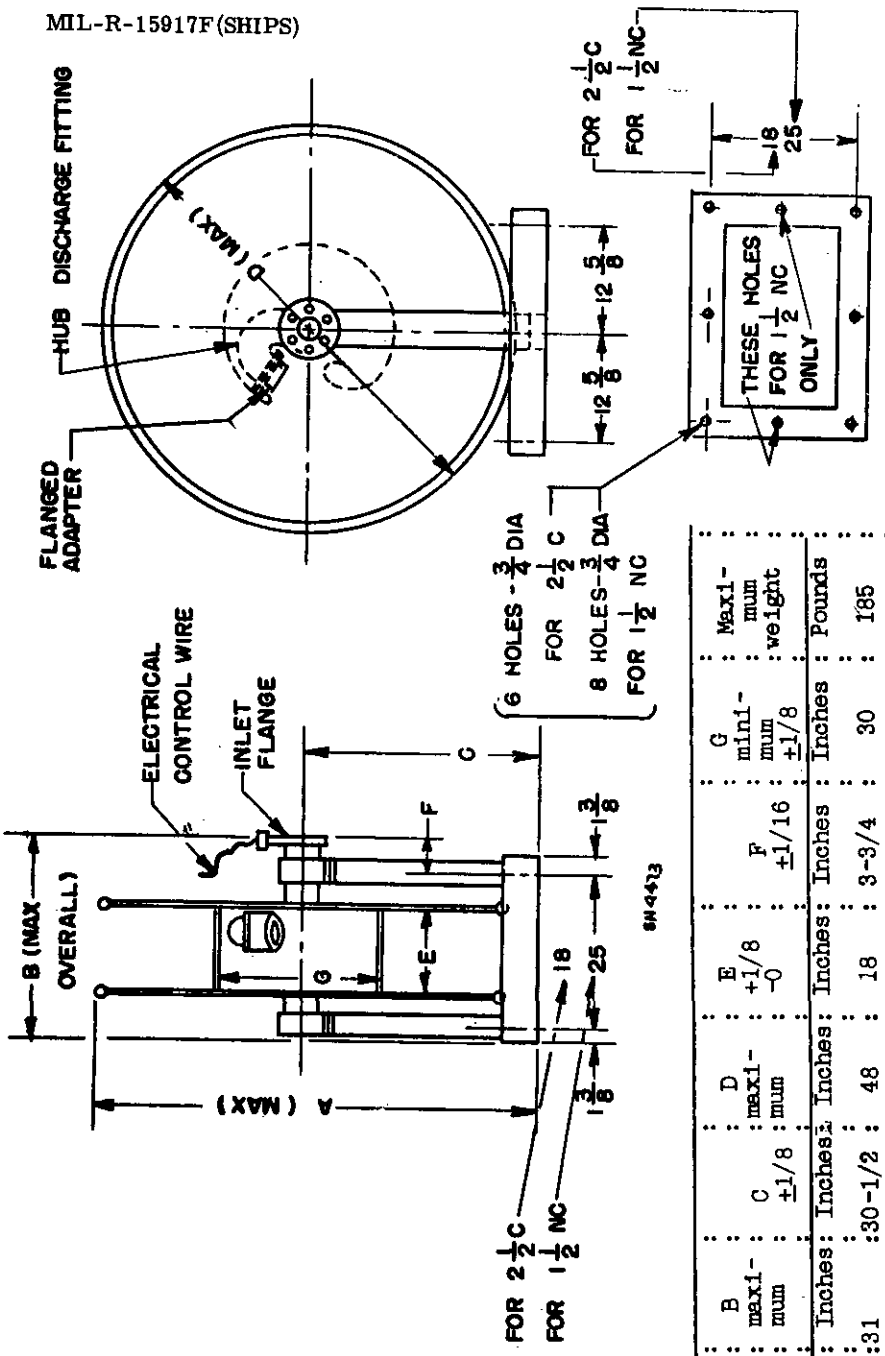
(a) Title, number, and date of this specification.
(b) Type of hose reel required (see 1.2).
(c) Whether right or left hand reels are required (see 3.9).
(d) Whether any special inlet flange requirements are required (see 3.12).
(e) Drawings required (see 3.13).
(f) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL - 15917, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Preparing activity:
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Hose reel type	A maximum	B maximum	C ±1/8	D maximum	E +1/8 -0	F ±1/16	G minimum ±1/8	Maximum weight
	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Pounds
Type II and III-1-1/2 inches NC	55	31	30-1/2	48	18	3-3/4	30	185
Type I and III-2-1/2 inches C	55	23-1/8	30-1/2	48	11	3-3/4	20	175

Figure 1 - Hose reel dimensions.

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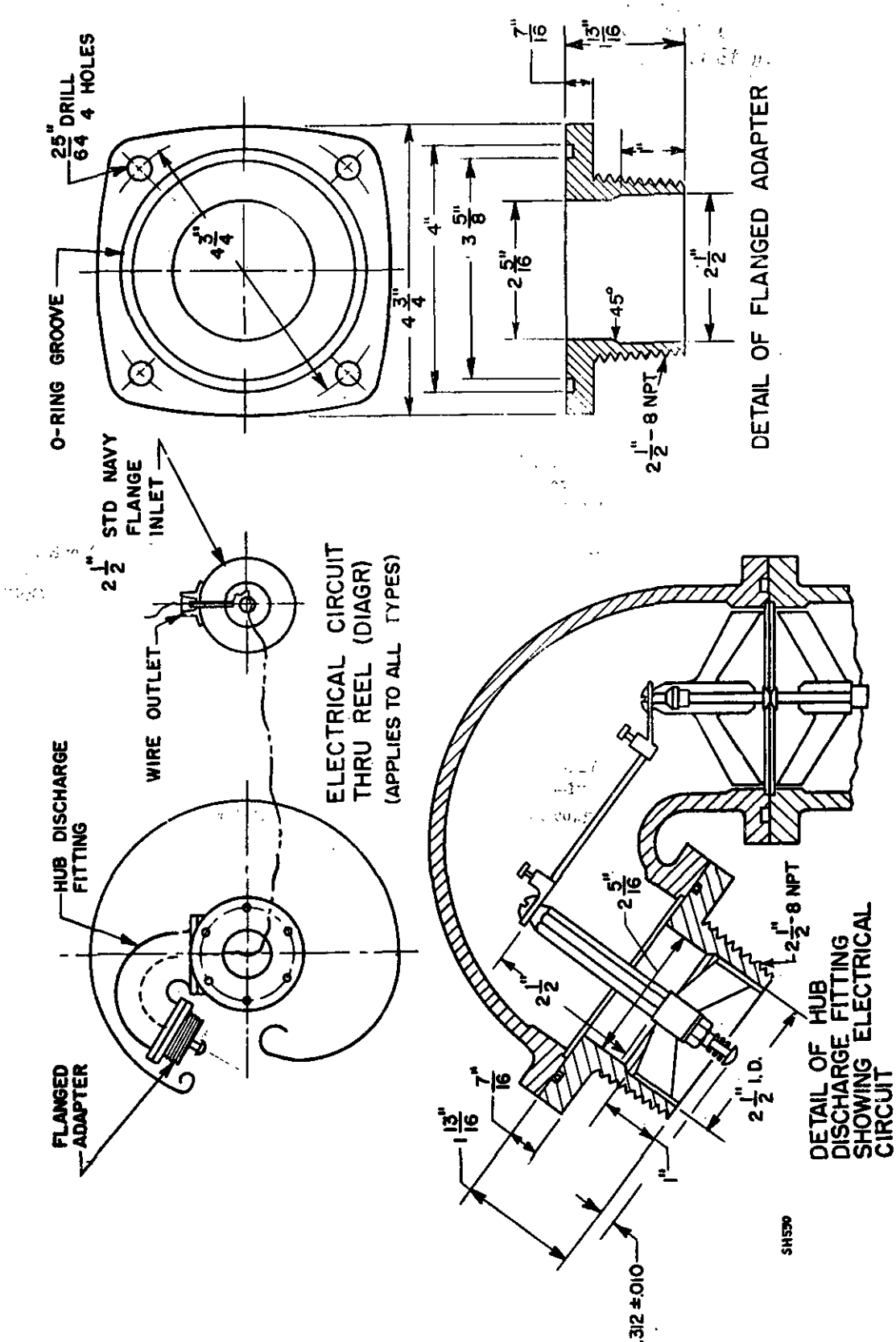


Figure 2 - Discharge fitting and adapter details for type I and III, 2-1/2 inch collapsible reels.

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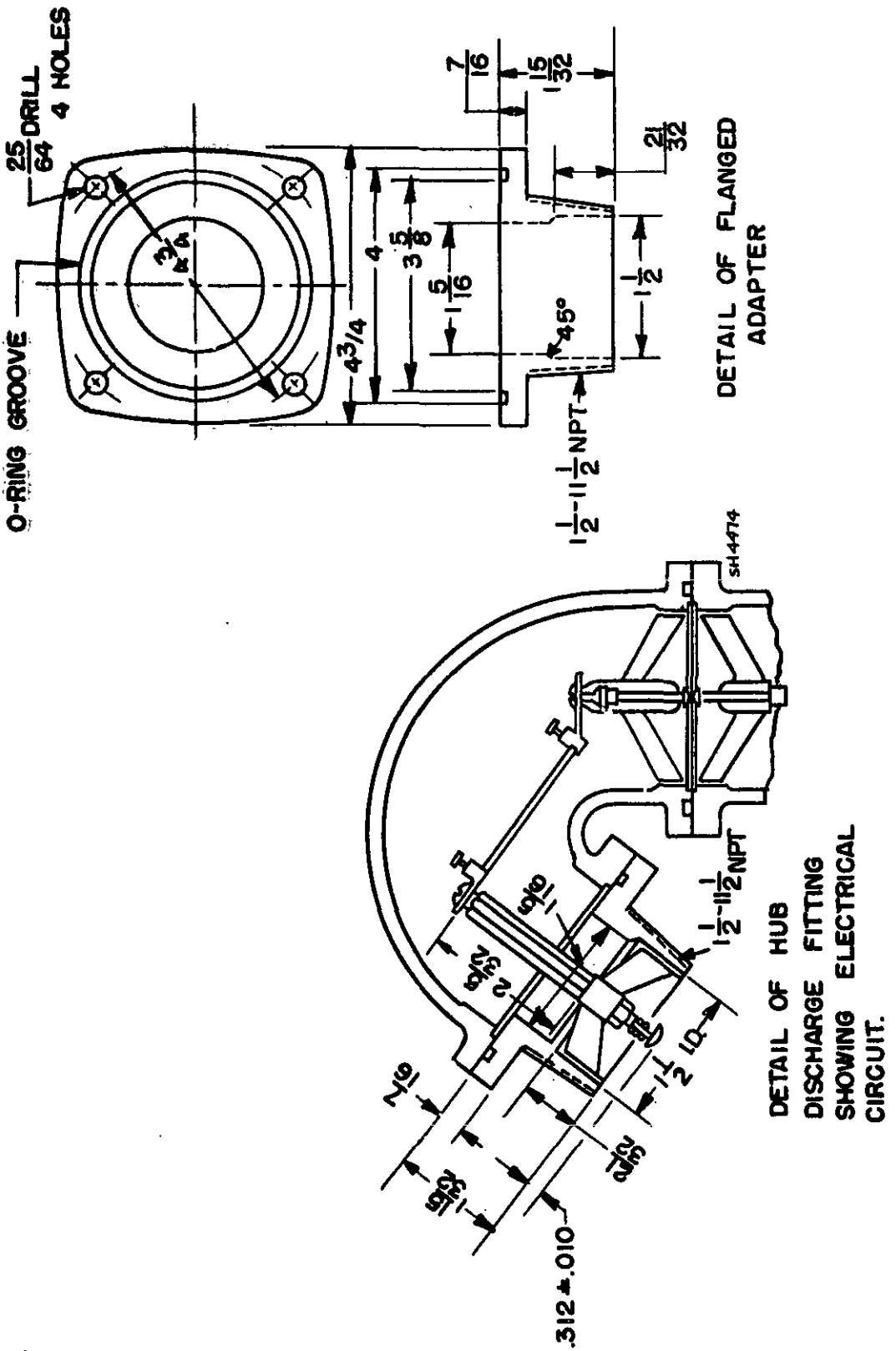


Figure 3 - Discharge fitting and adapter details for type II and III, 1-1/2 inch noncollapsible reels.

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